

2023-Ph-H1-Q10

Section: Electricity

Topic: Current, PD, Power, Resistance

A $3.0\ \Omega$ resistor has 12 C of charge pass through it in 4.0 s. What is the power?

Answer: E

(27 W)

Guidance for Students:

Use steps:

- Current $I = \frac{Q}{t} = \frac{12}{4} = 3\ \text{A}$
- Power $P = I^2 R = 3^2 \times 3 = 27\ \text{W}$

Revision Tips:

- Power formulas: $P = IV$, $P = I^2 R$, $P = \frac{V^2}{R}$
 - Use charge + time to find current.
 - Watch out for units!
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